

Mount / IK Fidelity Pilot

隔离机制试验: xArm6 / OpenArm, 官方模型与 TCP, 真实逐帧多分支 IK, 状态与扫掠碰撞检查。该报告不会改写正式十臂排名。

Artifact completeness: PASS

Strict full-episode results

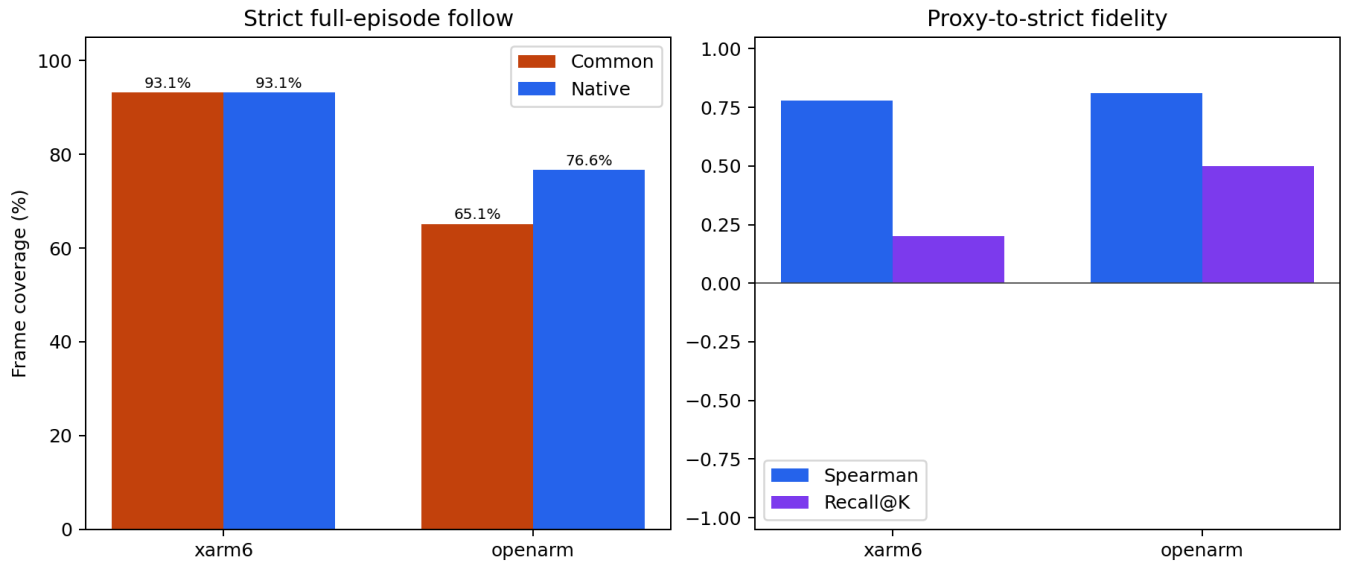
Robot	Common	Native	Common cov	Native cov	P95 pos	P95 ori	Spearman	Recall@K
xarm6	FAIL	FAIL	93.10%	93.10%	0.0055	0.0138	0.780	0.200
openarm	FAIL	FAIL	65.13%	76.63%	0.3413	0.4277	0.812	0.500

Winner mounts and failures

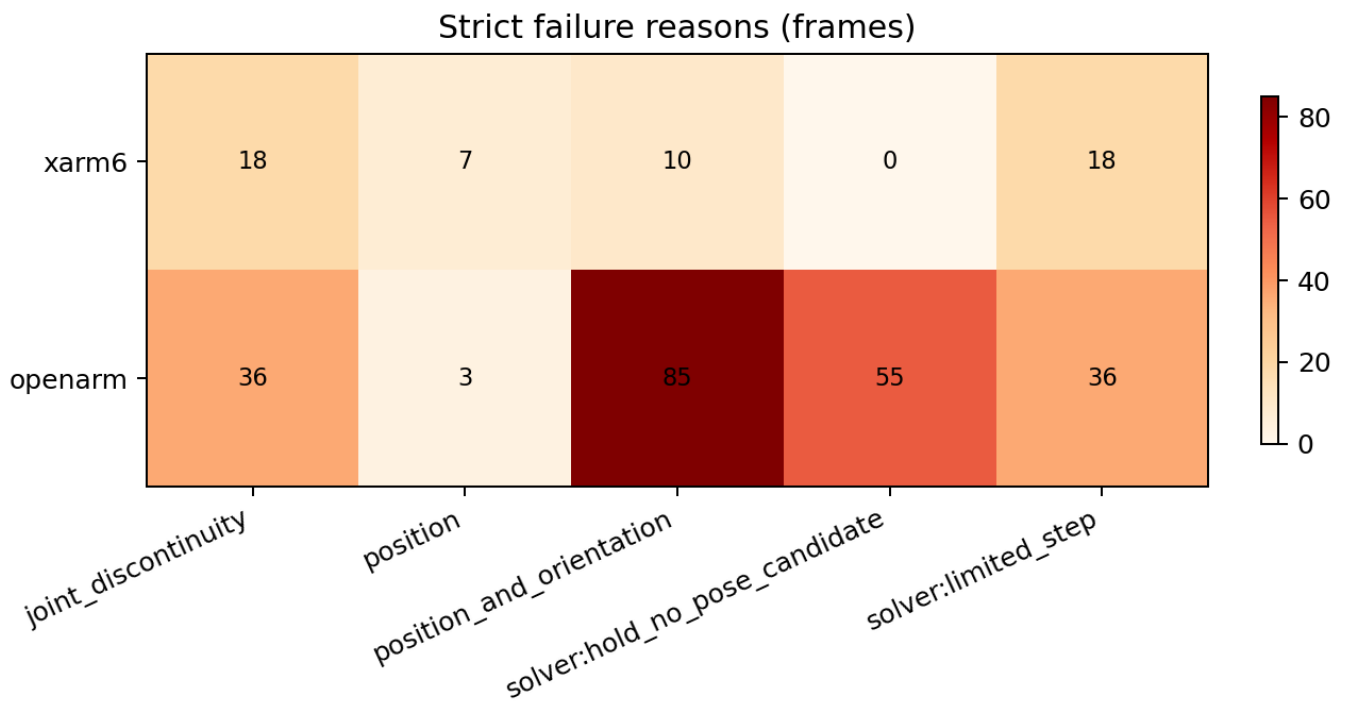
xarm6: X=0.213, Y=-0.550, Z=0.054 m; Tilt=0.0 deg, Yaw=-72.9 deg, Roll=0.0 deg. Failure frames: joint_discontinuity=18, position=7, position_and_orientation=10, solver:limited_step=18.

openarm: X=0.129, Y=-0.512, Z=0.251 m; Tilt=0.0 deg, Yaw=-144.5 deg, Roll=0.0 deg. Failure frames: joint_discontinuity=36, position=3, position_and_orientation=85, solver:hold_no_pose_candidate=55, solver:limited_step=36.

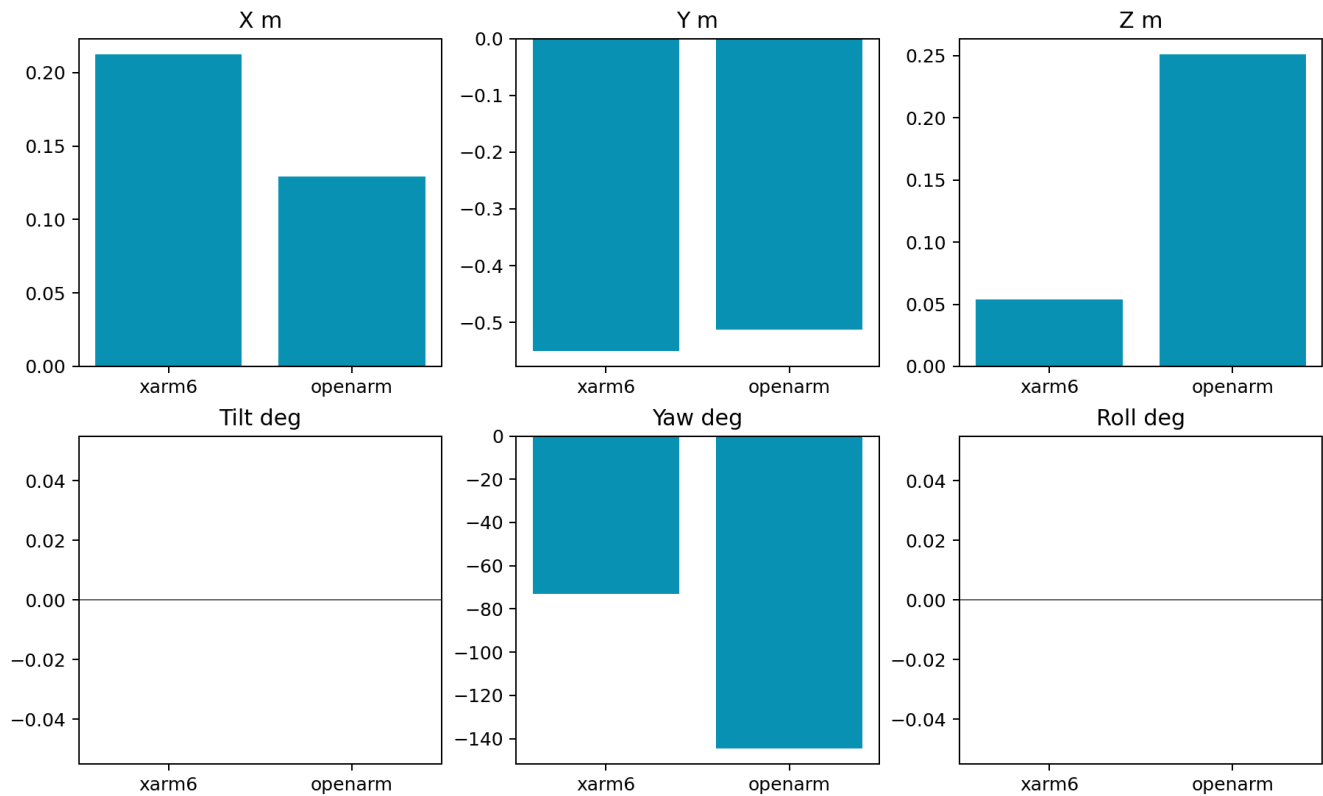
Outcome And Fidelity



Failure Reason Heatmap



Winner Mounts



Interpretation

The coarse capsule/proxy stage is a ranking and diversity mechanism only. Window screening cannot claim episode success. Final outcomes require every retained frame and transition to satisfy pose, continuity, velocity, joint-limit, state-collision, and swept-collision constraints.

Pilot conclusions

- Common-limit leader: xarm6; episode=FAIL, coverage=93.10%.
- Native-limit leader: xarm6; episode=FAIL, coverage=93.10%.
- xarm6 dominant failure evidence: joint_discontinuity (18 frames).
- openarm dominant failure evidence: position_and_orientation (85 frames).

Changes from the old flow

- All 4096 mounts participate in the diverse 256-candidate shortlist; proxy score is not a hard rejection gate.
- Strict windows and full episodes use the official model/TCP with real per-frame multi-branch IK.
- Path selection uses real timestamps, an effective finite horizon, state collision, and swept collision.
- Joint discontinuity is separated from pose failure and collision-blocked recovery.
- Common-limit and official native-limit finalists are evaluated separately; failures are rendered.